PY-STEM · ACTIVITY PYB-04 · COMPUTING

Barcode Checksum Rescue

Catch the fake barcode before the warehouse ships the wrong part.

Win condition: Most bad codes detected with few false alarms.

THE SCIENCE YOU ARE USING

Check digits catch errors. A 12-digit UPC-A code uses a weighted mod-10 check. Add the digits in odd positions 1, 3, 5, 7, 9, and 11 and multiply that sum by 3. Add the digits in even positions 2, 4, 6, 8, 10, and 12. A valid code makes the total a multiple of 10, so a single mistyped digit is rejected.

Detect without false alarms. A good detector flags the truly bad codes while passing the good ones. You race to find corrupted codes, keep false alarms low, and explain the check-digit rule you used.

HOW TO RUN IT

01 **Learn the rule.** For each 12-digit UPC-A code, multiply the odd-position sum by 3, add the even-position digits, and check whether the total is a multiple of 10.

02 **Verify a good code.** Apply the rule to a valid code and confirm it checks out.

03 **Hunt corrupted codes.** Scan a deck of cards and flag the ones that fail the rule.

04 **Avoid false alarms.** Double-check before flagging; wrongly rejecting a good code costs points.

05 **Explain the rule.** Describe how the check digit catches a single mistyped digit.

Safety: Normal classroom management. Allergy: None.

MATERIALS

Printed barcode cards	set
Dry erase boards	per team
Timers	per team

YOUR PLAN AND DATA

Show the weighted UPC-A arithmetic before accepting or rejecting a code.

Card number and 12-digit UPC-A code: ______________________

Odd-position sum x 3: ______________________

Even-position sum and grand total:

Verdict: accept or reject: __________

DEFEND YOUR VERDICT

Show the arithmetic for one rejected code. __________

Why does any single changed digit make this weighted total fail? __________

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	**100**